GGG-D-351F
May 6, 1987
SUPERSEDING
GGG-D-351E
July 8, 1966

FEDERAL SPECIFICATION

DIVIDER, MECHANICS'

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dividers used by machinists and sheet metal workers for scribing lines and laying out work.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. This specification covers the following types and classes when applicable, of dividers, as specified (see 6.2):

Type A. Spring.
  Class 1. Round legs.
  Class 2. Rectangular legs.
Type B. Firm joint.
Type C. Wing.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-C-320 - Chromium Plating (Electrodeposited).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For.
PPP-P-40 - Packaging and Packing of Hand Tools

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services Administration, Specifications Unit (WPSIS), 7th and D Street SW, Washington, DC 20407.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications.

MIL-C-13924 - Coating, Oxide, Black, for Ferrous Metals.
MIL-P-16232 - Phosphate Coatings, Heavy, Manganese or Zinc Base (For Ferrous Metals).

FSC 5210

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of dividers otherwise conforming to this specification.

3.2 Material. Divider legs (one-piece), and the point section of the two-piece divider legs shall be of tool steel, heat treated to produce tools capable of passing tests specified in 4.4.1.

3.3 Hardness. The points of divider legs shall be hardened. Dividers shall be capable of passing the practical use test specified in 4.4.1.

3.4 Identification marking. Each divider shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined, part number, and country of origin.

3.5 Finish. Dividers shall have a natural polished finish in accordance with the current tool industry practice or may be smoothly finished completely or in part with one or a combination of any of the following coatings at the option of the manufacturer:

(a) Chemically produced oxide coating, oil treated.
(b) Chemically produced phosphate coating, oil treated.
(c) Chrome, zinc, or equally suitable plating.

3.5.1 Coatings. The coatings shall be adherent, smooth, continuous, and free from pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability.

3.5.1.1 Bright chromium plate. The coating shall be electrodeposited metals consisting of nickel followed by chromium, the minimum thickness being 0.0002 inch for nickel, and 0.00001 inch for chromium. Plating adherence and thickness shall be tested in accordance with the test requirements for class 1 plating of QQ-C-320.

3.5.1.2 Dull or Matte chromium plate. Coating shall be supplied to surfaces sandblasted or otherwise treated before plating. The chromium coating shall have a minimum thickness of 0.0001 inch and shall be applied directly on the steel.

3.5.1.3 Flash chromium plate. The chromium coating shall have a minimum thickness of 0.00005 inch and shall be applied directly on the steel.

3.5.1.4 Hot phosphating. The coating shall be in accordance with MIL-P-16232, type M, class 2 or type Z, class 2.

3.5.1.5 Oxide coating. The coating shall be in accordance with MIL-C-13924, class 1, with a coating of preservative oil.

3.5.1.6 Zinc plate. The coating shall be in accordance with QQ-Z-325, type I, class 2.

3.6 Type A, spring dividers.

3.6.1 Class 1, round legs. The round leg spring dividers shall be similar to figure 1 and shall consist essentially of a circular band spring, two legs, nut, washer, threaded steel rod, thumb attachment, and fulcrum stud.

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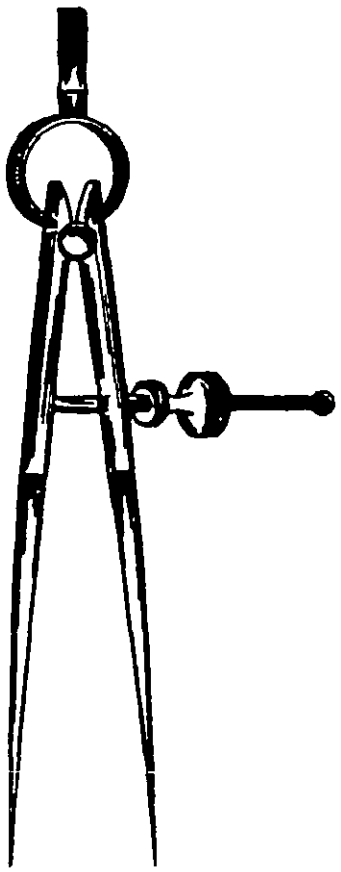


FIGURE 1. Type A, class 1, round leg spring dividers.

3.6.1.1 Circular steel band. Dividers shall have a hardened and tempered circular spring steel band at the hinged end, which shall cause the dividers to open and provide sufficient tension on the divider legs throughout their movement to hold them in a rigid manner. The band shall have a polished finish on the edges and outside surface.

3.6.1.2 Legs. Legs of dividers shall be straight, shall be round in cross section, and of sufficient size to minimize bending. The legs shall be tapered for at least half their length terminating in a sharp point. The points of the legs shall be equally distant from the fulcrum and shall come evenly together within 1/8 inch when the dividers are closed.

3.6.1.3 Nut. Dividers shall be adjustable by means of a steel nut and washer traveling over a finely threaded rod hinged to one leg of the divider and passing through the other leg. The nut shall be circular in shape, knurled on the periphery, and threaded to provide a neat fit on the threaded rod. Nuts shall be of solid one-piece construction.

3.6.1.4 Washer. A steel washer or other means shall be provided to reduce friction between the nut and leg or attachment on the leg. The bearing surfaces between the nut and washer and the washer and the leg or attachment shall be smoothly finished to facilitate ease of operation and reduce friction and wear.

3.6.1.5 Threaded rod. The steel adjusting rod shall have a thread as fine as the Unified National Fine series of thread. One end of the rod shall be hinged to one leg of the divider and the opposite end shall terminate in a small ball to prevent the nut from coming off. The rod shall be threaded for sufficient length so that when at their minimum range the legs can be adjusted to come in contact with each other, and so that the maximum working range shall be at least the size of the dividers.

3.6.1.6 Thumb attachment. A steel knurled finger and thumb attachment shall be provided. It shall be securely fastened to the circular steel band for convenience in handling the dividers when in use.

3.6.1.7 Fulcrum stud. The fulcrum stud shall be of steel, and shall provide a smooth bearing upon which the legs shall hinge. The fulcrum stud shall be designed to as to keep legs in alignment throughout the adjustment range.

3.6.1.8 Sizes. Sizes of dividers are defined as the distance from the center of fulcrum stud to the points of the legs. Type A dividers shall be furnished in 2-, 2-1/2-, 3-, 4-, 5-, 6- (± 1/16 inch) inch sizes; 8-, 10- (± 1/8 inch) inch sizes; or 12- (± 1/4 inch) inch sizes.

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3.6.1.9 Point protector. When specified (see 6.2), a point protector shall be provided. The point protector shall be of either metal-bound leather or plastic. When plastic protectors are furnished, they shall be of slow-burning type material. Point protectors shall be of a size most suitable for the size of dividers furnished.

3.6.2 Class 2, rectangular legs. The rectangular leg spring dividers shall be similar to figure 2 and conform to the requirements for class 1 dividers except that the legs shall be rectangularly in cross section, and the threaded rod shall be hinged to one leg and pass through an attachment on the other leg (see 3.6.1.3). When specified (see 6.2), a quick-adjusting nut shall be furnished for type A, class 2 dividers (see figure 2).

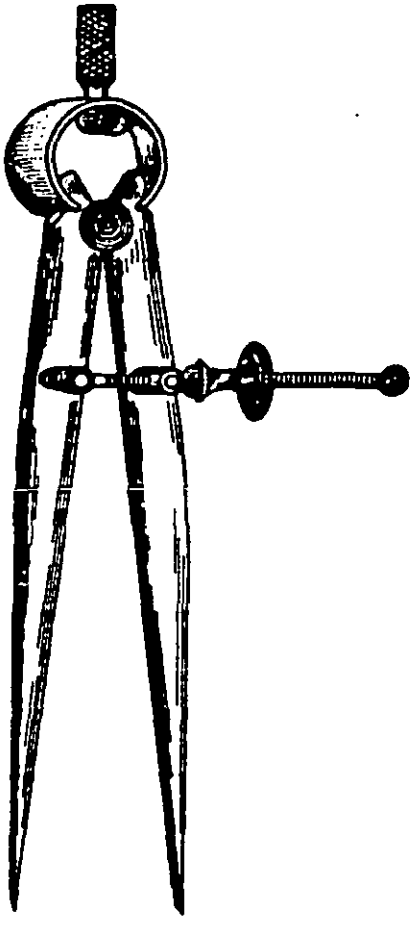


FIGURE 2. Type A, class 2, rectangular leg spring dividers.

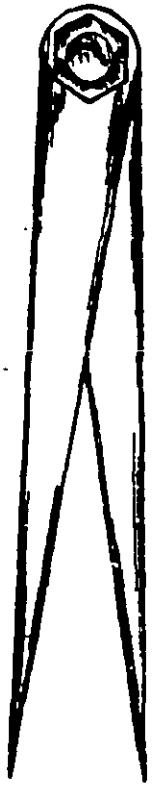


FIGURE 3. Type B, firm-joint dividers.

3.7 Type B, firm-joint dividers. The firm joint dividers shall be similar to figure 3 and consist of two legs jointed together at the pivot by means of a hexagonal or slotted-head adjusting screw.

3.7.1 Legs. Legs of type B dividers shall be of heat-treated tool steel. Legs shall be approximately parallel in thickness and tapered in width to terminate in sharply ground points. The legs shall come evenly together when dividers are in a closed position.

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3.7.2 Joint screw. An alloy steel joint screw shall be provided. This joint screw shall provide the pivot by which the legs are jointed together and shall be of such construction that it can be adjusted with a wrench (or screwdriver) to set and hold the legs to any desired tension or friction. The joint shall not show any backlash in any position of adjustment. Maximum capacity of dividers shall be at least equal to size of dividers.

3.7.3 Operation. All moving or sliding parts of dividers shall be constructed so as to provide smooth movement and operation.

3.7.4 Sizes. The firm joint dividers are designated in sizes which are defined as the distance between the center of the joint screw and the points of the legs. The dividers shall be supplied in 3-, 6-, or 12-inch sizes.

3.7.5 Point protector. When specified (see 6.2), a point protector shall be provided in accordance with 3.6.1.9.

3.8 Type C, wing dividers. The wing dividers shall be similar to figure 4 and consist essentially of two legs, an adjusting arc or wing, an adjusting spring, locking device, and an adjusting nut.

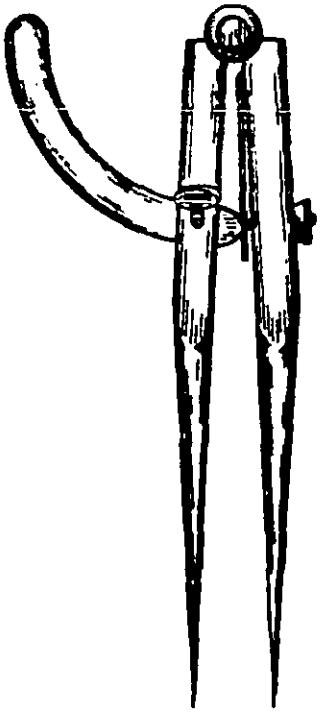


FIGURE 4. Type C, wing dividers.

3.8.1 Legs. Legs of type C dividers shall be of either one- or two-piece construction. When of one-piece construction, the entire leg shall be of tool steel. When of two-piece construction, the upper portion of the leg may be of pressed steel, and shall provide a means for permanently or securely attaching and holding the lower portion of the leg, which shall be of tool steel. When specified (see 6.2), the upper portion of the leg shall also accommodate a standard pencil of 5/16-inch diameter. Legs shall be square or generally rectangular on the upper section, and the lower section shall be round with a long taper ending in a sharp point. The point ends shall come evenly together when dividers are closed. When closed, the gap between the points shall be not more than 1/8 inch except for the 24-inch size which may be 3/8 inch. The legs shall be pivoted at the ends opposite the points. The pivot shall be a pin, rivet, or other equal means of attachment, neatly fitted with no side or end play.

3.8.2 Steel wing or adjusting arc. A wing or adjusting arc shall be provided. It shall pass through a slot in one leg of the divider. The opposite end shall have a threaded section which shall pass through an adjusting spring and a hole in the other leg. The wing shall be properly shaped to neatly fit the slot in the leg. The leg shall not catch or bind on the wing throughout the adjustment range. The wing or adjusting arc shall be of sufficient length for opening and adjusting the dividers to a minimum angle of 70 degrees. The find adjustment range measured at the points shall be not less than 3/8 inch.

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3.8.3 Adjusting spring. A spring-steel adjusting spring shall be affixed in such a manner to the leg through which the threaded section of the wing passes to provide sufficient tension to hold the leg rigid and allow for fine adjustments.

3.8.4 Locking device. A device for locking the leg on the wing-adjusting arc shall be provided to firmly lock the leg and wing at any position within the adjustment range. When a thumbscrew is used, the threaded section of the screw shall be of steel and not less than a number 6-32 machine screw size and shall engage not less than 2-1/2 threads of the divider legs.

3.8.5 Adjusting nut. The adjusting nut shall be of brass or steel, circular in shape and knurled on the periphery. The nut shall be threaded to neatly fit the threaded section of the wing or adjusting arc.

3.8.6 Operation. All moving or sliding parts of dividers of this type shall be so constructed as to provide smooth movement and operation.

3.8.7 Sizes. Type C wing dividers are designated in sizes which are defined as the distance between the center of the hinge pin and the points of the legs. Type C dividers shall be supplied in 6-, 7-, 8-, 9-, 10-, 12-, or 24-inch sizes.

3.9 Workmanship. Workmanship shall be first class throughout. The dividers shall be free from defects which may affect serviceability or durability. Burrs and sharp edges which might injure the operator shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements specified herein.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL-STD-105, except where otherwise indicated herein. Data for sampling shall be as stated in table I.

TABLE I. Sampling data.

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual and dimensional examination	1 ea.	II	Major 2.5 minor 4.0	Percent defective	4.3.1
Testing	1 ea.	S-4	2.5	Percent defective	4.4.1
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

4.3 Examination.

4.3.1 Visual and dimensional examination. Each sample unit shall be visually examined for the defects set forth below.

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TABLE II. Classification of defects.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Type, class, and size of dividers not as specified.
102	Divider material not as specified; evidence of cracks, laminations, rust, or other material imperfections.
103	Legs bent, warped, or not of sufficient cross section to minimize bending.
104	Points not sharp, not equally distant from the fulcrum, or points do not come together as specified.
105	Wing bent or crimped, does not provide the required adjustment angle, or wing sticks or binds in the divider leg (type C).
106	Adjusting spring defective, not securely attached to one leg, or spring fails to keep tension on the legs throughout the adjustment range (type C).
107	Locking device defective or fails to lock the leg to the wing at any position of adjustment (type C).
108	Fulcrum stud not secure or fails to keep the legs in alignment (type A).
109	Circular steel band not secure or fails to keep tension on the legs throughout the adjustment range (type A).
110	Adjusting rod not secure, fails to hinge on leg, does not furnish the required adjustment range; threads defective or not as required; threads stripped, torn, crossed, or not perfect full threads.
111	Adjusting nut not knurled, not furnished with a washer, threaded rod not terminated in a small ball (to prevent loss of the nut), or adjusting nut sticks, binds, or evidences excessive looseness when threaded the entire length of the rod (type A).
112	Dividers stick, bind, or evidence excessive looseness (type B).
113	Maximum working range of dividers less than the minimum required.
Minor:	
201	Dividers not finished as required; evidence of fins or burrs.
202	Length of tapered portion of legs not as specified (types A and C).
203	Threads of adjusting rod too coarse (type A).
204	Thumbscrew less than the specified size (type C).
205	Bearing surfaces of nut, washer, and leg not finished (type A).
206	Point protector (if required), not the required material or not snug fitting.
207	Marking, manufacturer's name or trademark missing, illegible, or not permanent.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1.

4.4.1 Practical-use test. Each sample divider shall be tested by setting at maximum capacity and scribing sufficient circles to equal a total perimeter of at least 24 feet. The plate on which the circles are scribed shall have a hardness value of not less than 30 nor more than 36 on the Rockwell C scale. An increase of 50 percent in the width of lines scribed or other evidence of point wear shall be evidence of test failure. Any divider in the sample failing the test shall be rejected. Each divider passing this test shall be considered as being properly hardened.

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4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2.1, 3.6.1.8, 3.7.4, and 3.8.7).
(c) Whether quick-adjusting nut is required (see 3.6.2).
(d) Whether point protector is required for types A and B (see 3.6.1.9 and 3.7.5).
(e) If leg shall clamp pencil (see 3.8.1).

MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an official interest in this Federal specification.

PREPARING ACTIVITY

GSA - FSS

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